

THE THIRD EYE GLOVE

An Arduino-Based Smart Obstacle Detection System

‘A Project for visually impaired community’

INTRODUCTION

- Visually impaired people face difficulty detecting nearby obstacles
- Traditional walking sticks have limited range and accuracy
- Technology can assist by providing real-time obstacle alerts
- **The Third Eye Glove** is a wearable, low-cost smart solution

What is it, actually?

Makes Me Aware

It helps visually impaired users understand the space around them without having to touch anything.

Finds Stuff

It uses these tiny ultrasonic sensors (like a bat!) to find objects that are nearby.

Gives a Buzz

It immediately changes the distance information into a vibration signal you can feel easily.

More Freedom

The main point is to help people walk around more safely, confidently, and all by themselves!

Problem Statement

- Lack of awareness about obstacles in front of visually impaired people
- Risk of accidents due to sudden obstacles
- Existing solutions are costly or inconvenient
- Need for a **portable, affordable, and easy-to-use device**

The Cool Design

- **Sensors are Smartly Placed:**

They are put on the glove in the best spots to see as much as possible.

- **Finger Signals:**

The vibrations happen on specific fingers, so you know exactly *where* the object is.

- **Distance is Clear:**

The buzz level clearly tells you:
“Danger, it’s right here!” or
“Relax, you have space.”

- **Fits Great:**

It’s shaped nicely to fit the hand, which makes it easy and comfortable to use.

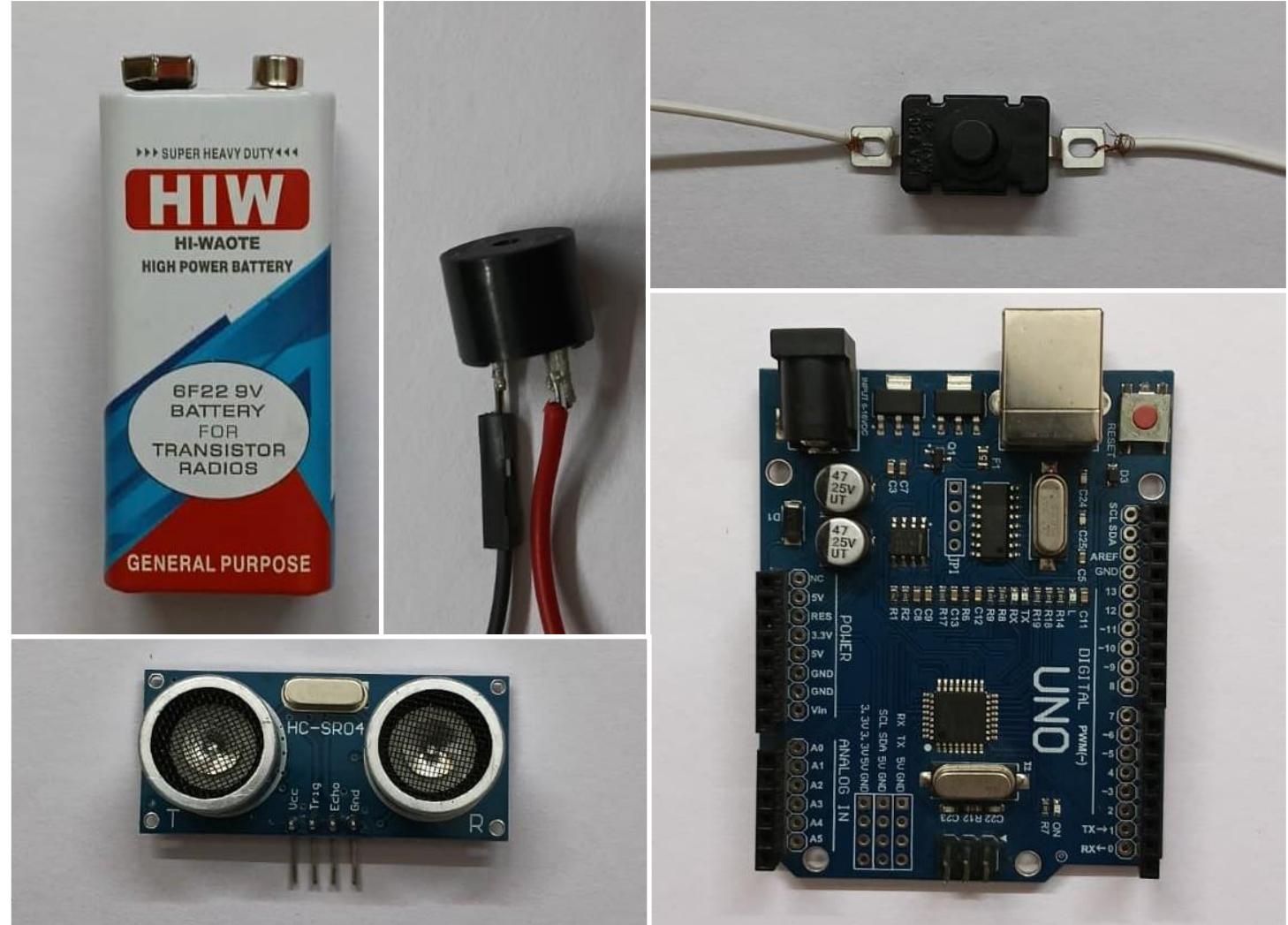


Objective of the Project

- To design a **smart wearable glove** for obstacle detection
- To detect obstacles using ultrasonic sensing
- To alert the user through sound/light signals
- To improve safety and independence of visually impaired users

Hardware Components Used

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Buzzer
- LED
- 9V Battery / Power Supply
- Switch
- Connecting Wires
- Glove (wearable base)



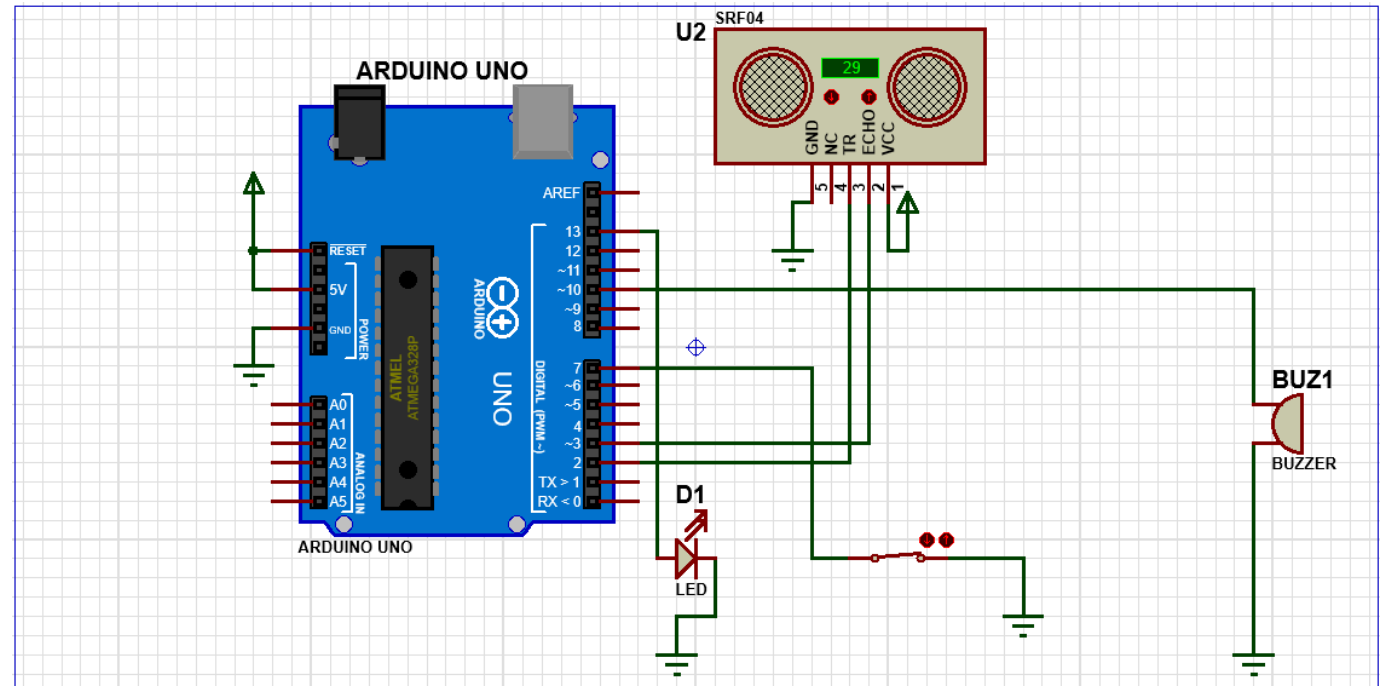
Software & Tools

- Arduino IDE (for coding)
- Embedded C / Arduino Programming
- Proteus (for simulation – optional)
- USB Cable for programming Arduino

Block Diagram

Blocks:

- Power Supply
- Ultrasonic Sensor
- Arduino UNO
- Output Devices (Buzzer & LED)



Working Principle

- Ultrasonic sensor emits sound waves
- Sound waves hit an obstacle and reflect back
- Arduino calculates distance using time delay
- If obstacle is within a set range:
 - . Buzzer alerts the user
 - . LED indicates proximity

How the Tech Works

- **It's Always Scanning:**

The sensors are always sending out sound waves to check the area in front of the user.

- **Buzz Tells Distance:**

If something is close, it buzzes harder. If it's far, it buzzes softly or not at all.

- **Super Fast Warning:**

It gives a real-time warning right on your hand—it's like an instant safety shield!

- **Easy to Wear:**

We designed it to be light and comfy so you can wear it all day without trouble.

Algorithm

- 1.Start the system
- 2.Trigger ultrasonic sensor
- 3.Measure echo time
- 4.Calculate distance
- 5.If distance < threshold:
 1. Turn ON buzzer
 - 2.Turn ON LED
- 6.Else:
 1. Keep outputs OFF
- 7.Repeat continuously

Advantages

- Low cost and portable
- Easy to use
- Real-time obstacle detection
- Wearable and hands-free
- Improves user safety

How It Helps People

- **Better Than the Cane:**

It's a modern upgrade—you can sense things without just poking around with a cane.

- **Simple Feelings:**

It takes complicated electronic data and turns it into simple, easy-to-understand feelings on your hand.

- **Walk with Confidence:**

This really helps people move, find their way, and feel much more confident about being out alone.

- **Stops Accidents:**

It's a constant safety net that drastically lowers the chance of bumping into things or falling.

Future Enhancements

- Add vibration motor for silent alerts
- Use GPS for navigation assistance
- Voice guidance using speaker module
- AI-based object recognition
- Mobile app integration

What I Want to Add Next!

Built-in GPS

I want to add outdoor tracking to help guide users to specific places.

Voice Helper

Maybe replace the buzzes with a voice telling them. things like "Stop" or "Turn Left"

AI Camera

Using a small camera and AI to recognize money, faces, or stop signs!

A Panic Button

A feature that could send an emergency text to family if the user falls.

Why I Call It the 'Third Eye'

- **Tech Combo:**

It smartly mixes the finding technology (ultrasound) with the feeling technology (vibration).

- **An Extra Sense:**

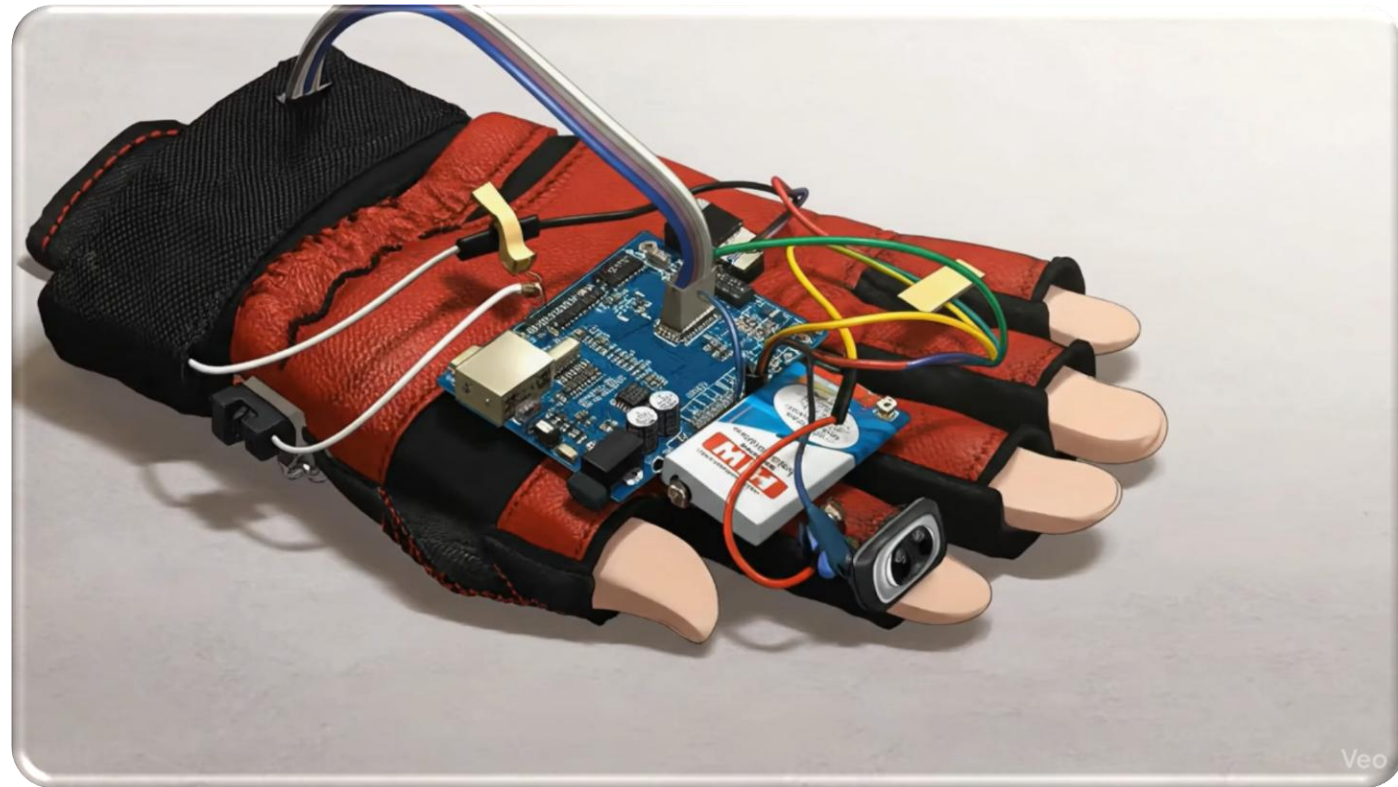
It gives the user an extra, artificial sense that they wouldn't normally have.

- **Sees Farther:**

It helps you know about objects that are beyond your reach—things you can't hear or touch yet.

- **Real Freedom:**

This ability to “see” beyond your hands is why it's so important for safe, independent living.



Real-World Uses

Every Single Day

It's great for everything—from walking in the hallway at home to crossing a busy sidewalk.

Works Anywhere

It keeps you safe in different places, both inside buildings and outside on the street.

Less Help Needed

It really reduces how much the user needs to rely on other people for directions.

Self-Esteem

It directly helps people feel better about themselves and their ability to be autonomous.

Focus: Getting Around

Instant Response

It finds and signals obstacles instantly—no waiting time at all.

Buzzing Maps

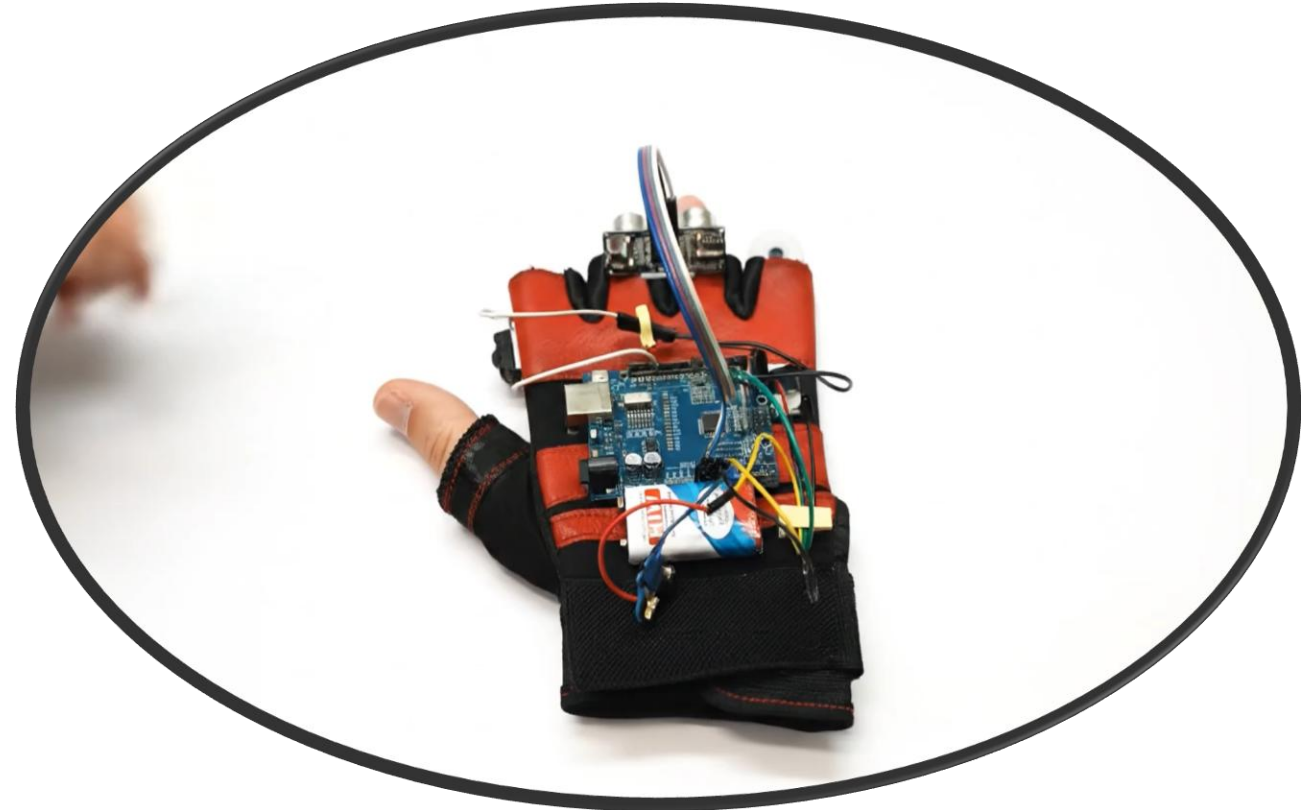
It tells you distance using patterns you can easily learn and remember.

Crowd Handler

It helps users move confidently even in crowded or tricky places like narrow doorways.

The Main Goal

Designed to help people move and travel completely on their own.



What's Inside? The Parts

The Brain (Arduino): This little computer chip takes all the data and makes all the decisions.

The Eyes (Ultrasonic Sensor): It's the part that sends out sound waves to 'see' obstacles.

The Sense (Buzzer): The buzzer is what buzzes to let the user know if they need to stop.

The Power (Battery): A small battery that makes sure the glove keeps running wirelessly.

How They Talk (The Circuit)

Simple Chat: The sensor constantly tells the Arduino board how far away objects are.

Decision Time: If the object is too close (uh oh, danger!), the Arduino sends a signal to start the motor.

Smart Pins: One pin sends the signal out ("Trigger"), and the other listens for the sound to bounce back ("Echo")

Wrapping Up

Freedom is Key

This glove isn't just tech; it's about giving people dignity and freedom to explore.

Simple and Cheap

By keeping the technology basic, we can make it affordable and available to everyone who needs it.

New Way to Sense

We're fixing the missing sense of sight by boosting the sense of touch.

My Dream

To help create a world where not being able to see doesn't stop anyone from doing what they want!

Conclusion

- The Third Eye Glove is an effective assistive device
- Provides safety and independence to visually impaired users
- Simple, affordable, and reliable solution
- Demonstrates practical application of Arduino and sensors

THANK YOU